
Sampling Theorem

Demonstration via SAR-based ADC

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Experiment Report

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1 Objective

This experiment aims to validate the Nyquist-Shannon sampling theorem by comparing the inputs and reconstructed signal:

- Sample and hold circuit for signal acquisition
- SAR-topology analog-to-digital converter for quantization
- Resistive ladder DAC coupled with smoothing filter for signal recovery
- Spectral examination via FFT to confirm adequate sampling across varying input frequencies
- Performance assessment of ADC through quantization step analysis, linearity metrics, and static error characterization including offset deviation, gain inaccuracy, DNL, and INL parameters

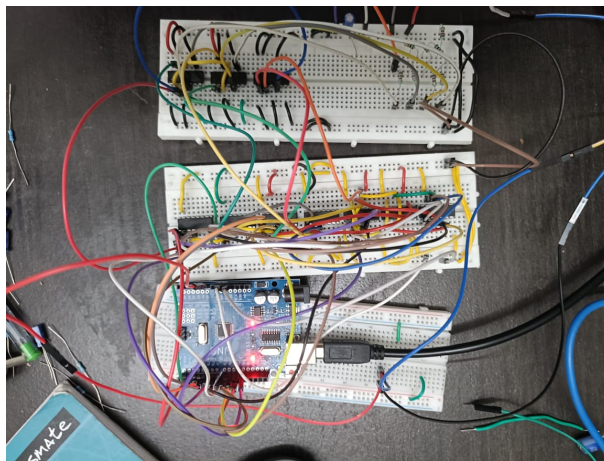


Figure 1: Circuit Picture

2 Theoretical Background

2.1 Nyquist-Shannon Sampling Criterion

The Nyquist-Shannon sampling criterion establishes that accurate reconstruction of a continuous-time waveform from discrete samples requires the sampling rate f_s to exceed twice the highest spectral component:

$$f_s \geq 2f_{max} \quad (1)$$

Here, f_{max} represents the maximum frequency present within the signal spectrum. The threshold sampling rate $f_{Nyquist} = 2f_{max}$ defines the Nyquist boundary.

2.2 Sample and Hold Circuit

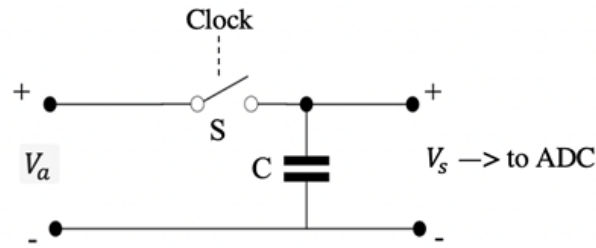


Figure 2: Sample and Hold Circuit

- A Sample and Hold (S/H) circuit is used to capture an instantaneous value of a continuous-time analog signal and maintain it constant for a short duration.
- The simple S/H circuit typically consists of an analog switch and a holding capacitor.
- During the **sampling phase**, the analog switch is closed, allowing the input signal to charge the holding capacitor. The capacitor voltage closely follows the input signal.
- During the **hold phase**, the switch is opened, isolating the capacitor from the input. The voltage stored on the capacitor is then held approximately constant and is applied to the input of the ADC. This held voltage remains stable for the duration of the conversion.

2.3 Successive Approximation Register ADC Principles

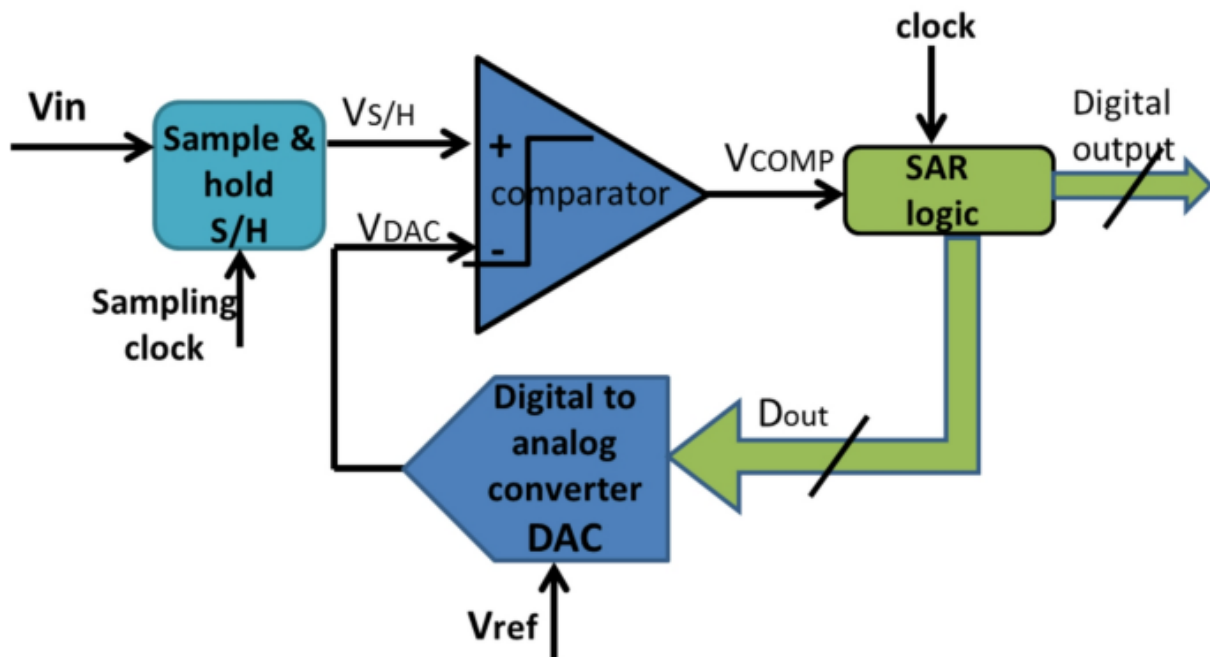


Figure 3: Block Diagram of the ADC

The SAR ADC topology employs a binary search methodology to map analog input voltages onto digital representations. The quantization procedure encompasses:

1. **Sampling:** The input voltage is captured and held constant during the entire conversion process.
2. **Bit-by-Bit Approximation:** The conversion starts by testing the most significant bit (MSB).
3. **Comparison:** An internal DAC generates a trial voltage, which is compared with the sampled

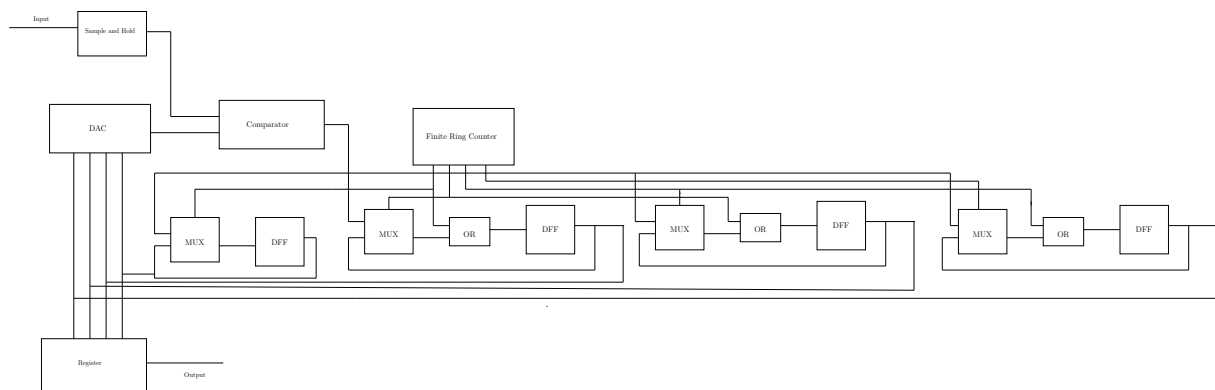
input voltage.

4. **Decision:** If the DAC voltage is lower than the input, the bit is set to 1; otherwise, it is set to 0.
5. **Iteration:** The same steps are repeated for each remaining bit down to the least significant bit (LSB).

An N-bit SAR converter necessitates N clock periods to complete one conversion cycle.

2.4 SAR ADC Implementation

The implemented SAR architecture uses custom logic built from cascaded multiplexer, OR gate, and flip-flop blocks to determine each bit sequentially. The overall circuit structure is shown below.



2.4.1 Timing Controller Behavior

The Finite Ring Counter serves as the timer, operating at the rapid clock frequency (10 kHz) while generating one-hot encoded control patterns:

Table 1: Ring Counter State Progression

Clock Cycle	FRC Output [4:0]
1	10000
2	01000
3	00100
4	00010
5	00001
6	00000
7	000000

An additional initial bit is included to synchronize the reset operation. When the second bit becomes logic high, it sets the D input of the first flip-flop, which is then updated on the following clock edge. This mechanism continues sequentially for the remaining bits.

2.4.2 Bit Change Mechanism

The SAR ADC determines the digital output using a binary search method. For a 4-bit SAR, the output code is represented as

$$B_3B_2B_1B_0$$

where B_3 is the most significant bit (MSB).

1. At the start of conversion, the SAR register is initialized with the MSB set to one:

$$\text{SAR} = 1000$$

This corresponds to a DAC voltage of

$$V_{DAC} = \frac{1}{2} V_{\text{ref}}$$

2. The DAC output is compared with the sampled input voltage V_{in} .
 - If $V_{in} \geq V_{DAC}$ (Comparator = 1), the MSB is retained and the next bit is tentatively set to 1:

$$\text{SAR} = 1100 \quad \Rightarrow \quad V_{DAC} = \frac{3}{4} V_{\text{ref}}$$

- If $V_{in} < V_{DAC}$ (Comparator = 0), the MSB is cleared and the next bit is set to 1:

$$\text{SAR} = 0100 \quad \Rightarrow \quad V_{DAC} = \frac{1}{4} V_{\text{ref}}$$

3. The same procedure is applied to all other bits.
4. After all four bits are tested, the final SAR register content represents the closest digital approximation of the input voltage:

$$D = \sum_{k=0}^3 B_k 2^k$$

2.4.3 Stable Output Capture Register

In a SAR ADC, the DAC output voltage fluctuates during the successive approximation process and settles only after all four clock cycles are completed. To ensure that only the settled digital value is captured, the final bit of the ring counter is routed through a delay block, which generates the clock signal for the final four output registers. When the delayed last-bit signal transitions from logic 0 to logic 1, the stable outputs of the SAR registers are sampled and stored.

2.5 R-2R Resistive Ladder DAC

The R-2R ladder DAC converts a digital input code into an analog voltage using only two resistor values, R and $2R$. Each digital bit controls a switch that connects the corresponding ladder node either to the reference voltage or ground.

Using superposition principle we get,

For an N -bit R-2R DAC, the voltage contribution of the k -th bit is given by:

$$V_{o,k} = B_k \cdot \frac{V_{\text{ref}}}{2^{k+1}}$$

where B_k is the digital value (0 or 1) of the bit.

By summing the contributions of all bits, the final output voltage becomes:

$$V_o = V_{\text{ref}} \sum_{k=0}^{N-1} \frac{B_k}{2^{k+1}} = V_{\text{ref}} \cdot \frac{D}{2^N}$$

where D is the decimal value of the digital input code.

Thus, the R-2R ladder DAC produces an analog output voltage proportional to the applied digital input.

2.6 Low Pass RC Filter

The reconstruction filter implements low-pass characteristics to attenuate high-frequency artifacts (sampling replicas) from DAC output. Ideal reconstruction demands cutoff frequency:

$$f_c = \frac{f_s}{2} \quad (2)$$

On using a capacitor of 1μ and resistor of $1K\Omega$ we get

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1000 \times 1 \times 10^{-6}} = 159.15 \text{ Hz} \quad (3)$$

This implementation employs RC topology with 159 Hz cutoff aligned to the Nyquist boundary.

2.7 ADC Metrics

2.7.1 LSB Voltage and Resolution

The LSB voltage defines minimum resolvable voltage increment:

$$V_{LSB} = \frac{V_{FS}}{2^N} \quad (4)$$

where V_{FS} denotes full-scale span and N specifies bit depth.

Resolution quantifies distinguishable output states: 2^N levels.

2.7.2 Offset Error

Offset deviation measures the discrepancy between actual initial code transition and theoretical threshold at $0.5 \times V_{LSB}$:

$$\text{Offset Error} = V_{measured}(code = 1) - 0.5 \times V_{LSB} \quad (5)$$

Sources include comparator bias, leakage currents, and reference inaccuracies.

2.7.3 Gain Error

Gain inaccuracy represents deviation of terminal transition from theoretical value:

$$\text{Gain Error} = Slope_{output} - Slope_{input}(\text{ramp stimulus}) \quad (6)$$

This affects transfer characteristic gradient, typically stemming from reference voltage errors or internal DAC imperfections.

2.7.4 Differential Nonlinearity

DNL quantifies each code width deviation from the ideal single LSB increment:

$$\text{DNL}(k) = \frac{V(k+1) - V(k)}{V_{LSB}} - 1 \quad (7)$$

where $V(k)$ represents transition voltage from code k to $k+1$. DNL helps in detection of missing codes.

2.7.5 Integral Nonlinearity

INL characterizes cumulative departure from ideal linear transfer function, representing total deviation between actual characteristic and ideal straight-line response:

$$\text{INL}(k) = \sum_{i=0}^k \text{DNL}(i) \quad (8)$$

3 Components

Table 2: Bill of Materials and Instrumentation

Item	Use	Qty
Arduino	Comparator	1
Resistors	1 k Ω (R-2R network, filter)	Multiple
Electronic Switch	For Sampling	1
Resistors	2 k Ω (R-2R network)	Multiple
Capacitor	1 μ F (RC filter and Sampling)	2
D-Type Flip-Flops	SAR Logic	Multiple
Ring Counter	Bit sequencing	1
Digital Oscilloscope	Digilent Waveforms platform	1
Bread Board	Circuit construction	3
Wire Jumpers	–	Multiple
DC Supply	+5V regulated	1

All measurements and spectral analysis are performed in the Digilent Waveforms oscilloscope platform.

4 Procedure

4.1 Hardware Assembly

1. Construct track-and-hold stage using analog switch IC and 1 μ F capacitor
2. Apply 320 Hz timing signal to switch control terminal
3. Implement SAR converter on bread board:
 - Configure ring counter for sequencing
 - Construct the SAR logic
 - Construct the second set for storing the stable value only.
 - Construct R-2R ladder using 1 k Ω and 2 k Ω resistors
 - Apply 10 kHz clock and the 320 Hz to SAR control logic
4. Implement output RC smoothing filter (1 k Ω , 1 μ F)
5. Position Waveforms oscilloscope probes for monitoring
6. Validate all interconnections and supply voltages

4.2 Nyquist Criterion Validation

1. Configure waveform generator for sinusoidal output at 20 Hz with amplitude from 0V to 5V.
2. Use the Scope in the platform to plot:
 - Input Wave
 - DAC output
 - Reconstructed Signal
3. Use the FFT tool for verifying.
4. Replicate procedure for 160 Hz and 260 Hz.

4.3 ADC Performance Evaluation

1. Apply gradual ramp stimulus spanning 0 to V_{FS} via waveform generator
2. Record input voltage at each code boundary using Waveforms platform
3. Use cursor functionality for accuracy.
4. Gather data across complete code range (0 through $2^N - 1$)

5 Experimental Data

5.1 Nyquist Criterion Confirmation

Table 3: Spectral Analysis Summary

Input	Expected Output	Observed Output	Remark
20 Hz	20 Hz	20 Hz	Faithful Result
160 Hz	160 Hz	160 Hz	Faithful Result
260 Hz	Folded to 60 Hz	~60 Hz	Aliasing confirmed

Given $f_s = 320$ Hz and $f_{Nyquist} = 160$ Hz:

- $20 \text{ Hz} < f_{Nyquist}$: Adequate sampling enables accurate reconstruction without spectral folding
- $160 \text{ Hz} \approx f_{Nyquist}$: Only 2 samples in 1 cycle making it look more distorted than last case.
- $260 \text{ Hz} > f_{Nyquist}$: Spectral folding manifests, signal appears at $|f_s - f_{input}| = |320 - 260| = 60$ Hz

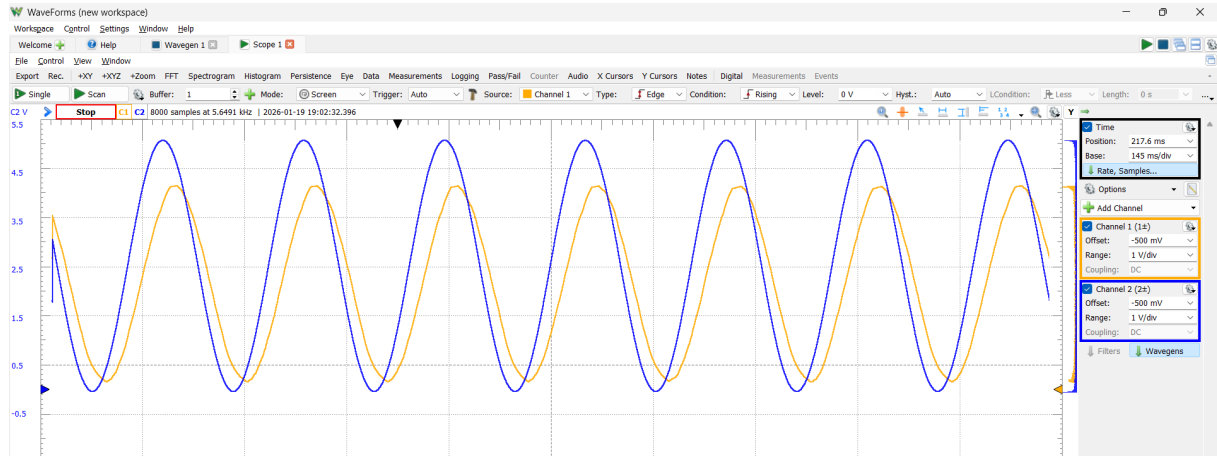


Figure 4: Input Frequency = 20Hz

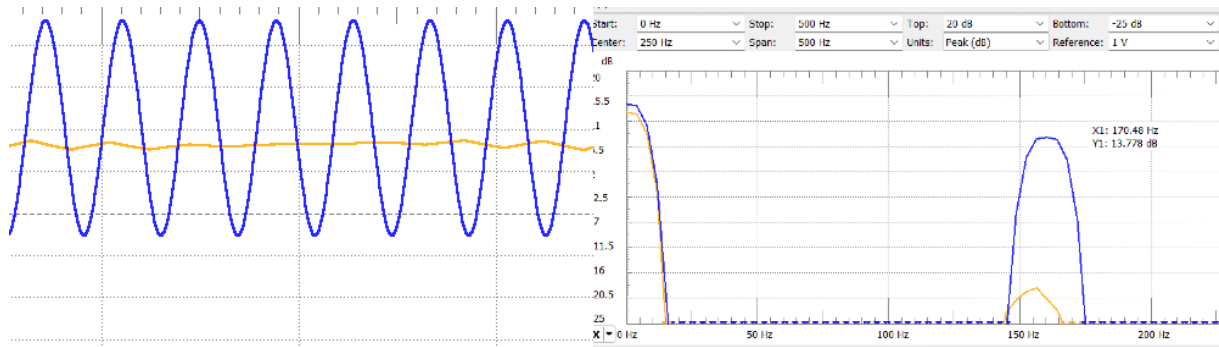


Figure 5: Input Frequency = 160Hz

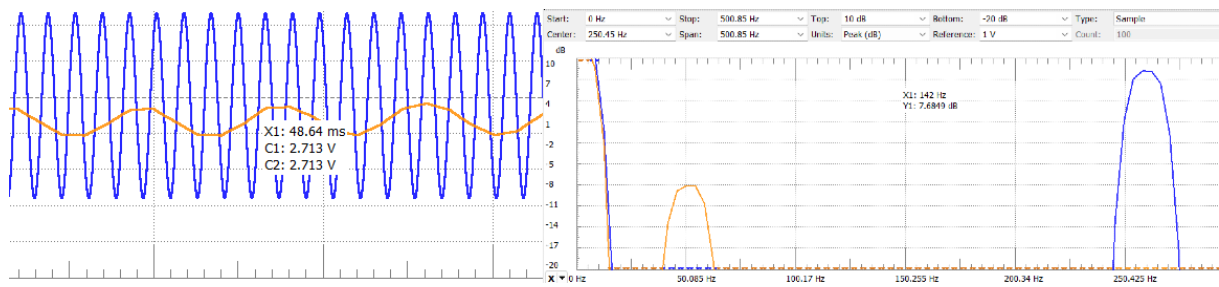


Figure 6: Input Frequency = 260Hz

6 Calculations

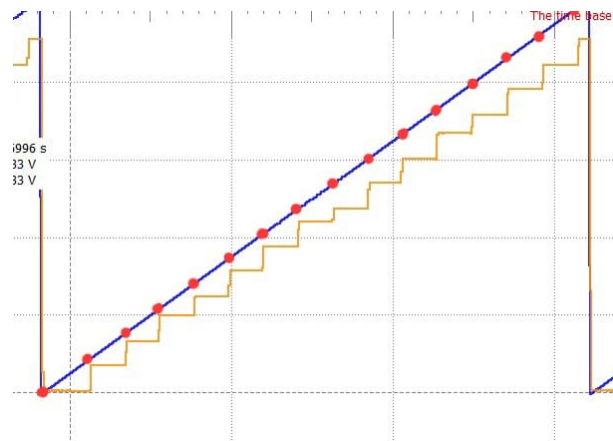


Figure 7: Slow Varying Ramp Input vs DAC output

The above figure was used to compute Offset Error , Gain Error, DNL and INL.

6.1 LSB Voltage and Resolution

For a 4-bit ADC with full-scale range $V_{FS} = 5V$:

$$V_{LSB} = \frac{5}{2^4} = 0.3125 \text{ V} \quad (9)$$

$$\text{Resolution} = 2^4 = 16 \text{ levels} \quad (10)$$

6.2 Conversion Time

$$\text{Conversion Time} = 4 \times \frac{1}{10^4} = 400 \mu s \quad (11)$$

6.3 Offset Error Calculation

$$\text{Offset Error} = 0.444 - 0.15625 = 0.28775 \text{ V} \quad (12)$$

$$\text{Offset Error (LSB)} = \frac{0.28775}{0.3125} = 0.92 \text{ LSB} \quad (13)$$

6.4 Gain Error Calculation

$$\text{Gain Error} = (5.048 - 5) - 0.28775 = -0.23975 \quad (14)$$

$$\text{Gain Error (LSB)} = \frac{\text{Gain Error (V)}}{V_{LSB}} = -0.7672 \quad (15)$$

6.5 DNL Calculation

For each code k :

$$\text{DNL}(k) = \frac{V_{measured}(k+1) - V_{measured}(k)}{V_{LSB}} - 1 \quad (16)$$

Table 4: DNL Calculation

Code	$V(k)$ (V)	DNL (LSB)
0	0.0000	+0.4208
1	0.444	+0.1104
2	0.791	+0.0304
3	1.113	-0.0528
4	1.409	+0.1552
5	1.77	-0.0528
6	2.066	-0.0496
7	2.362	+0.1104
8	2.709	+0.0720
9	3.044	+0.0304
10	3.366	+0.0272
11	3.687	+0.0304
12	4.009	+0.0307
13	4.331	-0.0336
14	4.633	+0.1734
15	5	-

$$\text{DNL}_{max} = 0.4208 \quad (17)$$

$$\text{DNL}_{min} = -0.0336 \quad (18)$$

6.6 INL Calculation

The INL was found by summing the DNL values till that specific bit.

$$\text{INL}(k) = \sum_{i=0}^k \text{DNL}(i) \quad (19)$$

Table 5: INL Calculation (Sample)

Code	INL (LSB)
0	0
1	+0.4208
2	+0.5312
3	+0.5616
4	+0.5088
5	+0.6640
6	+0.6112
7	+0.5584
8	+0.6688
9	+0.7408
10	+0.7712
11	+0.7984
12	+0.8288
13	+0.8592
14	+0.8256
15	+0.9990

$$\text{INL}_{max} = 0.9990 \quad (20)$$

$$\text{INL}_{min} = 0 \quad (21)$$

7 Results and Discussion

7.1 Nyquist Sampling Theorem Verification

The experimental results clearly demonstrate the Nyquist Sampling Theorem:

- **20 Hz** : It is below the the Nyquist frequency (160 Hz). The FFT analysis shows that the reconstructed signals contain the correct frequency components without aliasing. The time-domain waveforms closely match the input signals, confirming faithful reconstruction.
- **160 Hz signal**: This frequency is same as the Nyquist frequency, resulting in a slightly disoriented output. According to theory, the aliased frequency should appear at $f_s - f_{input} = 319 - 200 = 119$ Hz, which matches the measured FFT results. The time-domain signal appears as a lower frequency oscillation, demonstrating the consequence of undersampling.

- **260 Hz signal:** This frequency is above the Nyquist frequency by 100Hz. According to theory, the aliased frequency should appear at $f_s - f_{input} = 320 - 260 = 60$ Hz, which matches the measured FFT results. The time-domain signal appears as a lower frequency oscillation, demonstrating the consequence of undersampling.

This validates the fundamental principle: sampling must occur at least twice the maximum signal frequency to avoid aliasing and enable perfect reconstruction.

7.2 ADC Parameters

- LSB Voltage = 0.3125 V
- Resolution = 16 levels
- Offset Error : 0.92 LSB
- Gain Error : -0.7672 LSB
- DNL : -0.0336 LSB to 0.4208 LSB
- INL : 0 LSB to 0.9990 LSB
- Conversion Time : 400us

8 Conclusion

- This experiment successfully demonstrated the Nyquist–Shannon sampling theorem using a SAR-based analog-to-digital conversion system. Signals sampled below the Nyquist frequency were accurately reconstructed, while signals at and above the Nyquist limit exhibited distortion and aliasing consistent with theoretical predictions.
- The custom SAR ADC implementation operated correctly, completing conversion in four clock cycles with a measured conversion time of 400 μ s. Static performance evaluation showed no missing codes, with DNL and INL remaining within acceptable limits.
- Overall, the experiment validates both the theoretical principles of sampling and the practical operation of a SAR ADC, highlighting the importance of appropriate sampling frequency selection and illustrating key non-idealities present in real-world data converters.